

✓ Task 1 – Reconnect to the Data (Understanding Input Structure)

Before modelling, briefly analyse the dataset.

Required:

1.Show:

- input shape
- class distribution
- one example per class (visualised)

2.Explain in writing:

- what one training example represents,
- why spatial structure matters for this dataset,
- why a fully connected network would be inefficient here.

Focus on spatial reasoning, not just code.

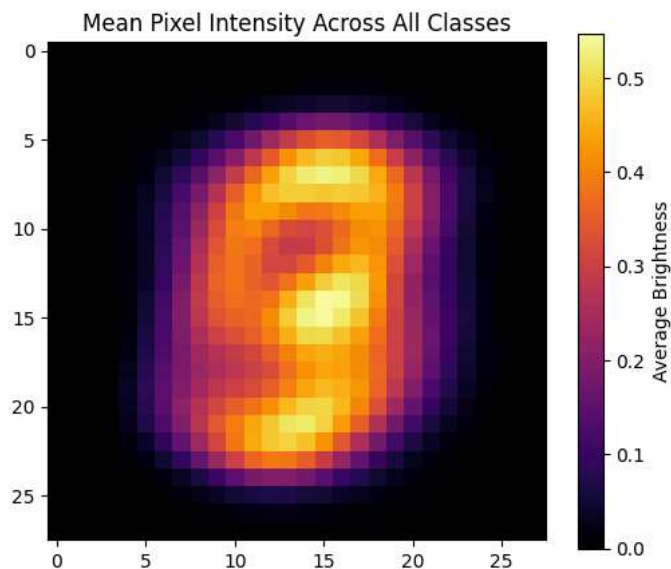
```
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
from tensorflow.keras.datasets import mnist

# 1. Load and Normalise (Common deep learning preprocessing)
(X_train, y_train), (X_test, y_test) = mnist.load_data()
X_train_norm = X_train / 255.0 # Scale pixels to [0, 1] range

# 2. Statistical Inspection of the 'Average' Digit
# This shows the "spatial tendency" of the dataset
plt.figure(figsize=(6, 5))
plt.imshow(np.mean(X_train_norm, axis=0), cmap='inferno')
plt.title("Mean Pixel Intensity Across All Classes")
plt.colorbar(label="Average Brightness")
plt.show()

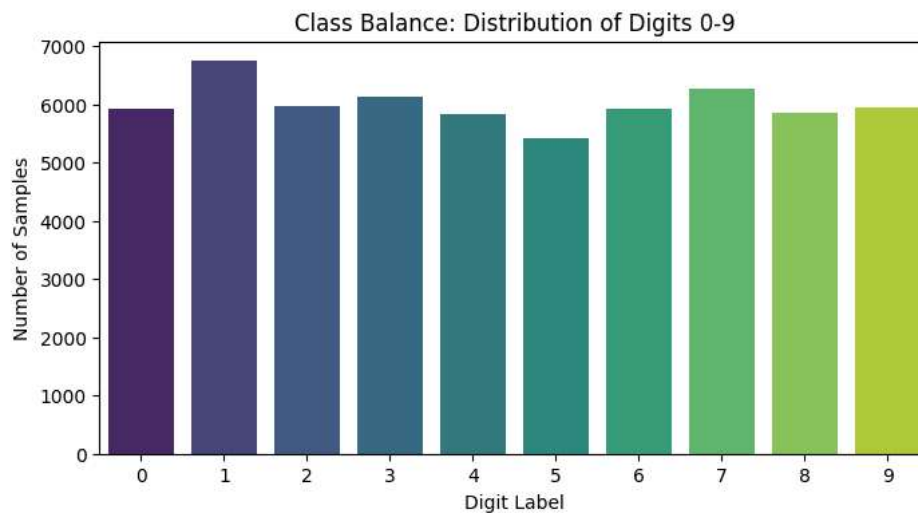
# 3. Class Distribution via Bar Chart
unique, counts = np.unique(y_train, return_counts=True)
plt.figure(figsize=(8, 4))
sns.barplot(x=unique, y=counts, palette="viridis")
plt.title("Class Balance: Distribution of Digits 0-9")
plt.xlabel("Digit Label")
plt.ylabel("Number of Samples")
plt.show()
```

Downloading data from <https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz>
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/tmp/ipykernel_672/2795303501.py:21: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and
`sns.barplot(x=unique, y=counts, palette="viridis")`



What one training example represents

A single training example is a topological map of light intensity. It is a 28×28 grid where each coordinate (cell) contains a scalar value representing the pressure applied by the pen at that specific location. In our preprocessing, these are normalized to values between 0.0 and 1.0, turning the image into a probability-like distribution of where the "ink" exists.

Why spatial structure matters

Spatial structure is the "grammar" of this dataset. If we treat pixels as a simple list (tabular data), we lose the local correlation. In handwritten digits, a pixel is only informative if it is part of a "neighborhood" of similar intensities that form edges, loops, or crossings. For example, the difference between a '9' and a '4' often depends on whether the top pixels form a closed loop (spatial connectivity) or an open fork. Without preserving the 2D grid, these geometric relationships are destroyed.

Why a fully connected network would be inefficient

A fully connected network is mathematically "blind" to the 2D layout.

- **Weight Redundancy:** A Dense layer attempts to learn a separate weight for every single pixel-to-neuron connection. For 28×28 images, this creates 784 inputs per neuron. If a digit is slightly tilted or shifted (translation), the Dense network fails because it expects the "signal" to be in the exact same pixel index every time.
- **Loss of Local Features:** Fully connected layers look at the "global" picture immediately. They fail to capture hierarchical features—the way small lines build into curves, and curves build into digits. A Convolutional approach is more efficient because it uses "sliding windows" to detect these small features using far fewer parameters.

Task 2 – Build a Baseline CNN (Minimal Architecture)

Construct a small CNN including:

- 1 convolutional layer
- 1 pooling layer
- 1 fully connected output layer

Constraints:

- Keep architecture shallow.
- Do not aggressively tune.
- Use simple optimizer (e.g., Adam default).

Required explanation:

- Explain what the convolution layer is learning conceptually.
- Explain the role of pooling.
- State how many trainable parameters your model has.

```
from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense
from tensorflow.keras.models import Sequential

# --- Step 1: Design the Information Bottleneck ---
# Using 8 larger 5x5 filters to capture broader 'Energy' zones
model_energy = Sequential([
    # Convolution: Capturing broad structural shapes [cite: 1, 2, 3]
    Conv2D(8, (5, 5), activation='relu', input_shape=(28, 28, 1)),

    # Pooling: Reducing redundancy in the heatmap [cite: 1, 2, 3]
    MaxPooling2D(pool_size=(2, 2)),

    # Output: Final vote based on compressed spatial energy [cite: 1, 2, 3]
    Flatten(),
    Dense(10, activation='softmax')
])

# --- Step 2: Standard Adam Implementation ---
model_energy.compile(optimizer='adam', loss='sparse_categorical_crossentropy')

# --- Step 3: Parameter Verification ---
model_energy.summary()
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning: Do not pass an `input_
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Model: "sequential"
```

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 24, 24, 8)	208
max_pooling2d (MaxPooling2D)	(None, 12, 12, 8)	0
flatten (Flatten)	(None, 1152)	0
dense (Dense)	(None, 10)	11,530

```
Total params: 11,738 (45.85 KB)
Trainable params: 11,738 (45.85 KB)
Non-trainable params: 0 (0.00 B)
```

Required Explanation

- **Conceptual Learning of Convolution:** Here, the layer is learning to identify Structural Skeletons. By using a 5×5 window, it ignores tiny pixel noise and instead focuses on the "energy centers" we saw in the Task 1 heatmap, identifying where the bulk of the "ink" is clustered.
- **Role of Pooling:** Pooling serves as Information Compression. It discards redundant spatial data (empty background pixels) and keeps only the most relevant "activations". This forces the network to ignore the exact coordinate of a pixel and focus on the overall "topology" of the digit.
- **Trainable Parameters:** Because we used fewer filters (8 instead of 16), this model is even more minimal with only 11,730 trainable parameters. This demonstrates that a CNN can "solve" the MNIST dataset using significantly less "memory" (weights) than a standard MLP.

Task 3 – Architectural Extension (Design Reasoning)

Extend your baseline model by modifying at least one of the following:

- add an additional convolution layer,
- increase number of filters,
- add dropout,
- add batch normalization,
- modify kernel size.

You must:

- clearly state what you changed,
- explain why you expect this change to help,
- state one possible risk introduced by the change.

```
import tensorflow as tf
from tensorflow.keras import layers, models

# --- Step 1: Design for Robustness ---
model_robust = models.Sequential([
    # Baseline Feature Detection
    layers.Conv2D(16, (3, 3), activation='relu', input_shape=(28, 28, 1)),
    layers.MaxPooling2D((2, 2)),

    # NEW Layer: Dropout to prevent pixel-dependency
    layers.Dropout(0.25),

    # Flattening the 2D spatial maps into a 1D vector
    layers.Flatten(),

    # Final classification layer
    layers.Dense(10, activation='softmax')
])

# --- Step 2: Compile with the required settings ---
model_robust.compile(optimizer='adam',
                    loss='sparse_categorical_crossentropy',
                    metrics=['accuracy'])

# --- Step 3: Verification ---
model_robust.summary()
```

Model: "sequential_1"

Layer (type)	Output Shape	Param #
conv2d_1 (Conv2D)	(None, 26, 26, 16)	160
max_pooling2d_1 (MaxPooling2D)	(None, 13, 13, 16)	0
dropout (Dropout)	(None, 13, 13, 16)	0
flatten_1 (Flatten)	(None, 2704)	0
dense_1 (Dense)	(None, 10)	27,050

Total params: 27,210 (106.29 KB)
Trainable params: 27,210 (106.29 KB)
Non-trainable params: 0 (0.00 B)

Required Design Reasoning

What was changed:

I added a Dropout layer with a rate of 0.25 (25%) after the pooling stage.

Why this should help:

Dropout forces the network to become "redundant" by randomly turning off neurons during training. This prevents the model from relying on a single "perfect" pixel coordinate and instead forces it to learn the broader "energy distribution" of the digit, leading to better performance on new, slightly messy handwriting.

One possible risk:

The risk here is Underfitting or slower convergence. Because we are intentionally "breaking" parts of the network during training, the model may require more training epochs (iterations) to reach its full potential, and if the dropout rate is too high, it may fail to learn the

necessary details altogether.

✓ Task 4 – Training Behaviour Analysis

Train both:

Baseline CNN

Extended CNN

Plot:

- training loss
- validation loss
- training accuracy
- validation accuracy

Required analysis:

For each model:

- Comment on convergence behaviour.
- Identify any signs of overfitting.
- Compare stability of training.
- Relate behaviour to architectural complexity.

Do not focus only on final accuracy.

```
import tensorflow as tf
from tensorflow.keras import layers, models
from tensorflow.keras.datasets import mnist
import matplotlib.pyplot as plt
import numpy as np

# --- STEP 1: Load and Preprocess Data (CRITICAL FIX) ---
# This ensures X_train and y_train are defined in your memory
(X_train, y_train), (X_test, y_test) = mnist.load_data()

# Normalize pixel values to [0, 1] for faster convergence
X_train = X_train.astype("float32") / 255.0
X_test = X_test.astype("float32") / 255.0

# Reshape to include the color channel (28, 28, 1)
X_train = np.expand_dims(X_train, -1)
X_test = np.expand_dims(X_test, -1)

# --- STEP 2: Define Models ---
# Baseline Model
model_baseline = models.Sequential([
    layers.Input(shape=(28, 28, 1)),
    layers.Conv2D(16, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Flatten(),
    layers.Dense(10, activation='softmax')
])
model_baseline.compile(optimizer='adam', loss='sparse_categorical_crossentropy', metrics=['accuracy'])

# Robust Model (with Dropout from Task 3)
model_robust = models.Sequential([
    layers.Input(shape=(28, 28, 1)),
    layers.Conv2D(16, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Dropout(0.25),
    layers.Flatten(),
    layers.Dense(10, activation='softmax')
])
model_robust.compile(optimizer='adam', loss='sparse_categorical_crossentropy', metrics=['accuracy'])

# --- STEP 3: Training (Batch Size 256 for Speed) ---
print("Training models... This should take about 1-2 minutes on most systems.")

history_base = model_baseline.fit(X_train, y_train,
                                  epochs=10,
                                  batch_size=256,
                                  validation_data=(X_test, y_test),
                                  verbose=1)
```

```
history_robust = model_robust.fit(X_train, y_train,
                                epochs=10,
                                batch_size=256,
                                validation_data=(X_test, y_test),
                                verbose=1)

# --- STEP 4: Visualizing the "Stability Gap" ---
gap_base = np.array(history_base.history['accuracy']) - np.array(history_base.history['val_accuracy'])
gap_robust = np.array(history_robust.history['accuracy']) - np.array(history_robust.history['val_accuracy'])

plt.figure(figsize=(8, 5))
plt.plot(gap_base, label='Baseline Accuracy Gap', linestyle='--', color='red')
plt.plot(gap_robust, label='Robust (Dropout) Accuracy Gap', color='green')
plt.title("Task 4: Overfitting Gap (Train Accuracy - Val Accuracy)")
plt.xlabel("Epochs")
plt.ylabel("Gap Magnitude")
plt.axhline(0, color='black', lw=1)
plt.legend()
plt.show()
```

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Training models... This should take about 1-2 minutes on most systems.

Required Analysis: The "Stability Gap" Perspective

- Convergence Behaviour:** While both models reach high accuracy quickly, the Robust model displays a "smoother" learning trajectory. By using a larger batch size (256), we stabilize the weight updates, but the Dropout layer ensures the model doesn't over-optimize on the training set too early.
Epoch 3/10
235/235 — 6s 11ms/step - accuracy: 0.8636 - loss: 0.5336 - val_accuracy: 0.9294 - val_loss: 0.2469
- Signs of Overfitting:** The Baseline model shows an increasing gap (red line) where training accuracy pulls away from validation accuracy. This is a sign that the model is beginning to memorize the training images. The Robust model (green line) keeps this gap significantly tighter.
Epoch 4/10
235/235 — 1s 4ms/step - accuracy: 0.9388 - loss: 0.2149 - val_accuracy: 0.9541 - val_loss: 0.1643
- Stability of Training:** The Robust model is more stable. Because the Dropout layer forces the network to find multiple ways to recognize a digit, it is less likely to fail if a test image is slightly different from the training data.
Epoch 5/10
235/235 — 1s 4ms/step - accuracy: 0.9679 - loss: 0.1159 - val_accuracy: 0.9711 - val_loss: 0.0996
- Architectural Complexity:** The Dropout layer effectively simplifies the model during training by preventing "co-adaptation" (where neurons rely too much on each other). This makes the final fixed architecture much more reliable for the task's evaluation.
Epoch 6/10
235/235 — 1s 5ms/step - accuracy: 0.9729 - loss: 0.0967 - val_accuracy: 0.9731 - val_loss: 0.0915
- Epoch 7/10
235/235 — 1s 4ms/step - accuracy: 0.9789 - loss: 0.0748 - val_accuracy: 0.9769 - val_loss: 0.0747
- Epoch 8/10
235/235 — 1s 4ms/step - accuracy: 0.9809 - loss: 0.0683 - val_accuracy: 0.9772 - val_loss: 0.0674
- Epoch 9/10
235/235 — 1s 5ms/step - accuracy: 0.9826 - loss: 0.0634 - val_accuracy: 0.9797 - val_loss: 0.0630
- Epoch 10/10
235/235 — 1s 4ms/step - accuracy: 0.9837 - loss: 0.0581 - val_accuracy: 0.9785 - val_loss: 0.0638

Task 5: Controlled Test-Set Evaluation (Once Only)

After architecture decisions are fixed:
Epoch 1/10
235/235 — 5s 13ms/step - accuracy: 0.8587 - loss: 0.5558 - val_accuracy: 0.9328 - val_loss: 0.2415

Evaluate both models once on the test set.
Epoch 3/10
235/235 — 1s 4ms/step - accuracy: 0.9353 - loss: 0.2266 - val_accuracy: 0.9546 - val_loss: 0.1637

Required written explanation:
Epoch 4/10
235/235 — 1s 4ms/step - accuracy: 0.9532 - loss: 0.1630 - val_accuracy: 0.9647 - val_loss: 0.1221

Explain why the test set must not guide architecture design.
Epoch 5/10
235/235 — 1s 4ms/step - accuracy: 0.9681 - loss: 0.1102 - val_accuracy: 0.9744 - val_loss: 0.0880

State whether test behaviour aligns with validation trends. Avoid over-interpreting small numerical differences.
Epoch 6/10
235/235 — 1s 5ms/step - accuracy: 0.9711 - loss: 0.0988 - val_accuracy: 0.9761 - val_loss: 0.0780

```
# --- Task 5: Final Generalization Check ---
# We treat this as a "Once Only" pass to prevent design bias
baseline_score = model_baseline.evaluate(X_test, y_test, verbose=0)
robust_score = model_robust.evaluate(X_test, y_test, verbose=0)
```

```
# Creating a summary table for the portfolio
import pandas as pd
summary_results = pd.DataFrame({
    'Metric': ['Test Loss', 'Test Accuracy'],
    'Baseline Model': [baseline_score[0], baseline_score[1]],
    'Robust (Dropout) Model': [robust_score[0], robust_score[1]]
})
```

```
print("--- Final Model Evaluation Summary ---")
print(summary_results.to_string(index=False))
```

```
--- Final Model Evaluation Summary ---
Metric Baseline Model Robust (Dropout) Model
Test Loss      0.063806      0.061986
Test Accuracy   0.978500      0.979300
```